School of Information Technology

IT1125 / IT1225 / IT1525 / IT1825 / IT1925
Web Development Project

Project Guidebook
AY2025 S2

1. Introduction

1.1 Aims

This module provides students with the practical experience of applying the concepts of object-oriented programming to develop software components that are not only maintainable and extensible but also aligned with sustainability principles. Students shall work in teams to develop, test and implement innovative and interactive applications aimed at solving real life problems that combines Design Thinking Methodology and Agile Processes.

1.2 Learning Outcome

At the end of this module, students will be able to:

- Explain and apply the concepts of classes, encapsulation, inheritance, and polymorphism in object-oriented programming in software development, with a focus on creating robust solutions.
- Apply the key concepts of object-oriented programming to construct maintainable, extensible, and scalable software applications.
- Apply design thinking approaches to create innovative and human-centric applications that effectively address user needs and requirements.
- Plan and execute a project as a team iteratively and deliver the solutions in a timely manner according to the given requirements, following established software development practices.

2. Project Scenario

2.1 Background

Singapore Chinese Chamber of Commerce and Industry (SCCCI) membership companies are looking for AI solutions to boost their productivity / effectiveness. Industry sectors include Finance & Insurance, Non-Profit Organisation, Manufacturing, Wholesale Trade and Construction.

2.2 Problem Statements

Select from the provided problem statements in **Batch 25S2 Problem Statement for AI PoC non SCCCI.pdf**, which includes the following sections:

Sections	Descriptions	Examples
Industry Sector	The type of sector	Retail & E-commerce
Problem Statement Title	A summary of the challenge presented by the company	AI-Assisted Product Search & Recommendation Tool for Customers
Who	The customer of certain service or product, or job function, never a company / department	Website visitors or potential customers looking for a product that suits their specific needs.
What	In-depth details on the issue	Currently, customers have to manually browse through categories or use

		basic search functions that often return too many or irrelevant results. There is no guided search experience that helps them filter and narrow down products based on their unique preferences or use cases. This increases drop-off rates and reduces conversions.
Barriers	Obstacles to user's journey or experience	Lack of intelligent filtering or guided search on the website. Too many products shown without relevance to customer intent. No recommendation engine to suggest best-fit options. Hard for non-technical users to identify the right product.
Cause	The underlying reason or factor that created the barriers	Existing website search is keyword-based only, not intent-driven. Product data not structured for advanced filtering or AI matching. No AI layer to rank or recommend based on context or user needs.
Emotion	The type of emotion that the problem has caused to the users	Frustrated, overwhelmed by choice, unsure which product to choose; fear of buying the wrong item.
Outcome of problem	The objective of solving the problem statement	High bounce rate due to customers not finding what they need quickly. Lower conversion rate as customers abandon the site without completing a purchase. Increased burden on customer support to assist with product selection.
Value proposition you hope AI can bring about	What type of values that AI can provide for solving the problem statements	AI-assisted search tool that guides customers through a few smart filters (e.g., usage, budget, specs) and returns the top 3–5 best-fit product recommendations. Helps customers make confident, faster decisions. Improve conversion rates by 20–40% through personalized product discovery. Reduce reliance on human support for basic product guidance.
Does this project potentially require your company data?	Company data will be provided if there is a requirement for using them	Yes – includes product specs and current website structure.
What company system(s) would this PoC likely	The requirement of whether there is a need to work with existing system(s) that the	Website CMS or e-commerce platform Product catalog or inventory database

need to work with, if any?	company already has in production	

Recommended problem statements:

- 1A – Automating Job Card Update
- 1B – Automate quotation
- 2C – Product recommendation tool
- 2E – Marketplace for IT services
- 3A – Demand forecasting
- 3B – Procurement optimization
- 3C – Pricing optimization
- 4A – Job fit scoring
- 6A – Group buy management
- 6B – AI for sales and marketing
- 6C – AI for order-to-fulfilment process

Project teams may choose other problem statements not listed above and/or develop multiple problem statements from the same company. Project teams are required to choose up to three problem statements and at least one problem statement will be selected for development.

2.3 Project Requirements

To design the web application with a focus on meeting the needs of the target users. Innovation and thinking beyond existing services are encouraged to create a unique and valuable user experience. Each project team to consist of **4 to 5 students**. Each student is responsible for developing at least one feature. Each feature must include core functions (create, retrieve, update, delete (CRUD) operations).

Each team can come up with the basic features such as, but not limited to:

- Accounts Management (login/logout, profile, user management, session management)
- Products Management (product catalogue, product details, product management)
- Orders (shopping cart, order summary, payment, order management)
- Reviews and Rewards (reward management, payment, product review management)
- Customer Support (customer support system, report, dashboard)

2.4 Minimum Feature Requirements

- Must fulfil the requirements set by each problem statement and the supporting core functions.
- Must utilise Python classes and objects in the web application.

- Must utilise Python Flask and JinJa2 frameworks.
- Must utilise SQLite or other forms of storage (e.g. MySQL) for implementing persistent storage for Python objects.
- Data validation to ensure data consistency must be performed.

2.5 Additional Features (where applicable)

- Requirements that are mentioned in the problem statements that falls under the following features are considered additional features:
  - Chatbot for customers to interact with.
  - Notifications that will update in real-time.
  - Exporting report in various Excel format.
  - Uploading function using Excel as template.
  - Real-time search and filter in listing records.
  - Map integration with location-based services or route planning.
  - Integration with third-party APIs or libraries.
  - Cloud hosting of web application.
- The features listed above are not exhaustive; additional recommended features can be considered to further enhance the solution.

2.6 Tools and Technologies

The application framework for this project shall utilise Flask for building web applications with Python. Where appropriate, you may utilise additional development tools, such as Bootstrap, to further enhance the functionality of your project. Teams following predefined problem statements should adhere to any specific technology requirements outlined in their chosen problem statement.

2.7 Project Timeline & Deliverables

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Week	Deliverables	Graded	Notes
1	• Form project teams • Submit 3 preferred problem statements • Attend mass briefing for the industry collaboration		• Teams of 4 to 5 students • Choose problem statements that match your interests
2	• Project problem statement allocated • Attend meeting with company for requirements		Ask questions to clarify requirements and take notes for planning your features
4	Discuss team task allocation with tutor		Decide who will develop which features
9	Proposal presentation	20%	First project assessment

13	Tutors to check in with students and nominate groups		Nominated groups will prepare for interim presentation
14	Interim presentation to company		Nominated students to present solutions to clients
15	Technical Review	20%	Second project assessment
18	Final presentation	40%	Final project assessment
19	Final check in by client nominated groups		Nominated students to present final solutions to clients

3. Project Assessment

Project comprises 80% of the total module mark.

3.1 Assessment Components

Week	Deliverables	Individual	Group	Total
9	Proposal Presentation			20%
	- Innovation		5%	
	- Proposal		5%	
	- Presentation	5%		
	- Prototype	5%		
15	Technical Review			20%
	- User Interface Usability		5%	
	- Flask Application	5%		
	- Technical & Functionality	5%		
	- Field Validations	5%		
18	Personal Portfolio Submissions			40%
	- Resume and Cover Letter	5%		
	- Digital Portfolio	5%		
	Final Presentation			
	- Integration		5%	
	- Final Solution	15%		
	- LinkedIn Sharing	5%		
	- Presentation	5%		
		60%	20%	80%

3.2 Assessment Rubrics

Week 9 Proposal Presentation (20%)

Group (10 Marks)					
Category	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Very Good (B)	Excellent (A)
Innovation Base Score: 5 Marks	0-1 business needs with little analysis, description, and relevant key facts of the business needs that require digital innovation by the company/industry. 0 to <2.5	1-2 business needs that require digital innovation by the company/industry, each with some detailed analysis, description, and relevant key facts. 2.5 to <3	2-3 business needs that require digital innovation by the company/industry, each with some detailed analysis, description, and relevant key facts. 3 to <3.5	3-4 business needs that require digital innovation by the company/industry, each with detailed analysis, description, and relevant key facts. 3.5 to <4	At least 4 business needs that require digital innovation by the company/industry, each with detailed analysis, description, and relevant key facts. 4 to 5
Proposal Base Score: 5 Marks	Proposed functionality exhibits little empathy for user and incorporated workflow that requires more than 5 improvements before it can value add to business needs. 0 to <2.5	Proposed functionality exhibits low level of empathy for user and incorporated workflow that requires 4 to 5 improvements before it can value add to business needs. 2.5 to <3	Proposed functionality exhibits good empathy for user and incorporated mostly feasible, realistic workflows that requires up to 3 improvements before it can value add to business needs. 3 to <3.5	Proposed functionality exhibits good empathy for user and incorporated mostly feasible, realistic workflows that adds good value to business needs with minimal improvements needed. 3.5 to <4	Proposed functionality exhibits high level of empathy for user and incorporated feasible, realistic, and practical workflow that value add to business needs. 4 to 5

Individual (10 Marks)					
Category	Inadequate	Acceptable	Satisfactory	Very Good	Excellent

	(F)	(D)	(C)	(B)	(A)
Presentation Base Score: 5 Marks	Presenter speaks with monotone; rarely in complete sentences and not easy to understand and follow; Presenter rarely maintains eye contact throughout the presentation. 0 to <2.5	Presenter speaks with adequate articulation, voice projection and confidence; Presenter occasionally maintains eye contact throughout the presentation. 2.5 to <3	Presenter speaks with strong articulation, voice projection and confidence; Presenter maintains eye contact most of the time throughout the presentation. 3 to <3.5	Presenter speaks with excellent articulation/volume and confidence; Presenter maintains eye contact throughout most of the presentation. 3.5 to <4	Presenter speaks with excellent articulation/volume and confidence; Presenter maintains eye contact throughout the presentation and demonstrates exceptional engagement with audience. 4 to 5
Prototype Base Score: 5 Marks	Wireframe design with limited details and little to no validation messages that requires more than 5 improvements in meeting user needs. 0 to <2.5	Wireframe design with details and validation messages that requires 3 to 5 improvements in meeting user needs. 2.5 to <3	Wireframe design with details and validation messages that requires up to 2 improvements in meeting user needs. 3 to <3.5	Wireframe design with complete details and validation messages that fully meets user needs with minor enhancements. 3.5 to <4	Wireframe design with comprehensive details, complete validation messages, and exceptional user experience considerations in meeting user needs. 4 to 5

<u>Week 15 Technical Review (20%)</u>

Group (5 Marks)					
Category	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Very Good (B)	Excellent (A)

User Interface Usability Base Score: 5 Marks	Use of bootstrap to create a UI with basic layout, navigation and fonts, and has minimal human-centric design and usability with significant inconsistencies. 0 to <2.5	Use of bootstrap from practical to create a UI with basic layout, navigation and fonts, and has minimal human-centric design and usability. 2.5 to <3	Use of bootstrap to create a UI with adequate consistency in layout, navigation and fonts, and has good human-centric design and usability. 3 to <3.5	Use of bootstrap to create a UI with a good consistency in layout, navigation and fonts, and has very good human-centric design and usability with responsive UI. 3.5 to <4	Use of bootstrap to create a UI with excellent consistency in layout, navigation and fonts, and has high human-centric design and usability with responsive UI. 4 to 5

Individual (15 Marks)					
Category	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Very Good (B)	Excellent (A)
Flask Application Base Score: 5 Marks	Completed less than 1 complete function (C, R, U or D) with significant errors that prevent proper operations. 0 to <2.5	Completed 1 function (C, R, U or D) with at least 2 fields and no errors. 2.5 to <3	Completed 2 functions (C, R, U and/or D) with at least 2 fields for each function and no errors. 3 to <3.5	Completed 3 functions (C, R, U and/or D) with at least 2 fields for each function and no errors. 3.5 to <4	Completed 4 or more functions (C, R, U and/or D) with at least 2 fields for each function and no errors. 4 to 5
Field Validations Base Score: 5 Marks	Basic validation rules implemented with minimal validation messages and limited coverage of fields.	Basic coverage of validation rules ensures data submitted by users is in the correct format for some fields and with minimal validation messages in meeting user needs.	Adequate coverage of validation rules ensures data submitted by users is in the correct format for most fields and with adequate validation messages to meet user needs.	Good coverage of validation rules ensures data submitted by users is in the correct format for most fields and with good validation messages to meet user needs.	Extensive coverage of validation rules ensures data submitted by users is in the correct format for all fields and with excellent validation messages to meet user needs.

	0 to <2.5	2.5 to <3	3 to <3.5	3.5 to <4	4 to 5
Technical and Functionality Base Score: 5 Marks	Basic implementation of input field types with limited consideration for business and user needs. 0 to <2.5	Basic use of relevant input field types across some fields to make the functionalities that meet the minimum needs of the business and the users. 2.5 to <3	Adequate use of relevant input field types across most fields to make the functionalities that meet the needs of the business and the users. 3 to <3.5	Good use of relevant input field types across most fields to make the functionalities effective in meeting the needs of the business and the users. 3.5 to <4	Excellent use of relevant input field types across all fields to make the functionalities easy and effective in meeting the needs of the business and the users. 4 to 5

Week 18 Personal Portfolio Submission (10%)

Individual (10 Marks)					
Category	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Very Good (B)	Excellent (A)
Resume and Cover Letter Base Score: 5 Marks	Resume is missing multiple sections that are essential. Resume appears cluttered and messy. Cover letter fails to highlight suitability.	Resume is missing 2 essential sections. Resume is formatted in an acceptable manner, but there are multiple inconsistencies. Cover letter mentions suitability but needs further elaboration.	Resume is missing 1 essential section. Resume is formatted in a neat manner but has a few inconsistencies. Cover letter mentions suitability but could be improved with elaboration.	Resume includes the essential sections. Resume is formatted in a neat manner. Cover letter highlights suitability.	Resume includes all essential sections. Resume is formatted in a professional and neat manner. Cover letter is well written, highlights strengths and suitability.

	0 to <2.5	2.5 to <3	3 to <3.5	3 to <3.5	4 to 5
Digital Portfolio Base Score: 5 Marks	Information provided on project sample is unclear and lacked any explanation. No use of visuals, graphics and videos. 0 to <2.5	Information provided on project sample is somewhat unclear and needs further explanation. Very little use of visuals, graphics and videos. 2.5 to <3	Information provided on project sample shows attempt at explanation. Some use of visuals, graphics and videos. 3 to <3.5	Information provided on project sample is clear and generally supported with explanation. Good use of visuals, graphics and videos. 3 to <3.5	Information provided on project sample is clear and well supported with explanation. Excellent use of visuals, graphics and videos. 4 to 5

Week 18 Final Presentation (30%)

Group (5 Marks)					
Category	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Very Good (B)	Excellent (A)
Integration Base Score: 5 Marks	Solution is poorly integrated with significant errors throughout and/or requires 3 or more members' solutions to run separately on different workstations 0 to <2.5	Solution is not fully integrated, runs with 6 or less errors and/or requires 2 members' solution to run separately on another workstation. 2.5 to <3	Solution is not fully integrated, runs with 4 or less errors and/or requires 1 member's solution to run separately on another workstation. 3 to <3.5	All members' solutions are fully integrated, and solution runs with 2 or less errors. 3.5 to <4	All members' solutions are fully integrated with no errors. 4 to 5

Individual (25 Marks)

Category	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Very Good (B)	Excellent (A)
Final Solution Base Score: 15 Marks	Basic solution with minimal or no additional features that shows limited understanding of user needs and business requirements. 0 to <5	Average solution with additional features implemented that exhibits little empathy for user and incorporated workflow that requires extensive enhancement before it can value add to business needs. 5 to <6	Good solution with additional features implemented that exhibits adequate empathy for user and incorporated somewhat feasible, realistic, and practical workflow that required substantial enhancement before it can value add to business needs. 6 to <7	Very good solution with additional features implemented that exhibits good empathy for user and incorporated mostly feasible, realistic and practical workflow that required minimum enhancement before it can value add to business needs. 7 to <8	Excellent solution with additional features implemented that exhibits high level of empathy for user and incorporated completely feasible, realistic, and practical workflow that adds significant value to business needs. 8 to 10
	Demonstration lacks proper structure, session management, and test data, making it difficult to follow the workflow. 0 to <2.5	Poorly structured demonstration of solution without session and little to no test data setup for demonstration of workflow between processes. 2.5 to <3	Adequately structured demonstration of solution with session and basic test data setup for demonstration of workflow between processes. 3 to <3.5	Well-structured demonstration of solution with session and some test data setup for smooth demonstration of workflow among processes. 3.5 to <4	Excellently structured demonstration of solution with session and relevant test data setup for smooth and neat demonstration of workflow among processes. 4 to 5
LinkedIn Sharing Base Score: 5 Marks	Unclear or incomplete description of the project's objectives with no highlighting of key features and no mention	Limited description of the project's objectives, little highlights of key features, and minimal outlining of benefits and	Adequate description of the project's objectives, highlighting some key features, and outlining	Well-defined description of the project's objectives, highlighting most key features, and outlining several	Clear and concise description of the project's objectives, extensively highlighting key features, and

	of benefits or positive impacts. 0 to <2.5	positive impacts of the solution. 2.5 to <3	few benefits and positive impacts of the solution. 3 to <3.5	benefits and positive impacts of the solution. 3.5 to <4	outlining all the benefits and positive impacts of the solution. 4 to 5
Presentation Base Score: 5 Marks	Presenter speaks with monotone; rarely in complete sentences and not easy to understand and follow; Presenter rarely maintains eye contact throughout the presentation. 0 to <2.5	Presenter speaks with adequate articulation, voice projection and confidence; Presenter occasionally maintains eye contact throughout the presentation. 2.5 to <3	Presenter speaks with strong articulation, voice projection and confidence; Presenter maintains eye contact most of the time throughout the presentation. 3 to <3.5	Presenter speaks with excellent articulation/volume and confidence; Presenter maintains eye contact throughout most of the presentation. 3.5 to <4	Presenter speaks with excellent articulation/volume and confidence; Presenter maintains eye contact throughout the presentation and demonstrates exceptional engagement with audience. 4 to 5

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